**20** Two bags, **A** and **B**, each contain only green marbles and red marbles.

There are 6 green marbles and 4 red marbles in bag **A**.
There are 8 green marbles and 2 red marbles in bag **B**.

One marble is going to be taken at random from bag **A** and placed in bag **B**.

A marble is then going to be taken at random from bag **B**.

Work out the probability that this marble will be a green marble.

If green is transferred, B has 9 green out of 11.

**1 mark**
B now has 11 marbles.

If red is transferred, B has 8 green out of 11.

$$P(\text{green from B}) = \frac{6}{10} \times \frac{9}{11}$$

**1 mark**
Multiply along paths.

$$+ \frac{4}{10} \times \frac{8}{11}$$

**1 mark**
Add both green paths.

$$= \frac{54 + 32}{110} = \frac{86}{110}$$

$$= \frac{43}{55}$$

**1 mark**
86/110 = 43/55.

.....

**(Total for Question 20 is 4 marks)**